



Financed Emissions Dashboard

User's Manual

First Bank of Nepal

Demonstration Build | v0.1 (Demo Build)
June 2026

DEMO PURPOSES ONLY

This document accompanies a non-production demonstration build of the Financed Emissions Dashboard prepared for First Bank of Nepal. The bank name, branding, and loan book are illustrative.

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About this demonstration

This product, including its branding and the bank identity used throughout, is a demonstration build prepared to illustrate how a financed emissions dashboard could integrate with a commercial bank in Nepal. It is not a regulatory filing tool, and the outputs shown in this manual are not suitable for use in NRB submissions, NSRS disclosure, audit, or external reporting.

What is real in this demonstration

The reference emissions data underneath the dashboard is drawn from public sources that the Jana platform has ingested and stabilised. Specifically, the dashboard references:

- **Climate TRACE v5.6 (2024)** facility-level CO₂-equivalent emissions for 213 Nepal assets, summing to approximately 28.04 Mt CO₂e.
- **EDGAR v8.1** gridded national CO₂, polygon-clipped to the Nepal administrative boundary, yielding 1,343 cells and an estimated 18.81 Mt national CO₂ for the latest available year.
- **Global Cement and Concrete Tracker (July 2025)** for 61 operating cement plants in Nepal, with capacity and owner metadata.
- **NRB ESRM Guidelines (2018)**, including Annex 5 ESDD checklist categories.
- **NRB Green Finance Taxonomy (October 2024)** categorisation of bank lending activities into green, amber, red and unclassified.
- **PCAF Global GHG Accounting and Reporting Standard**, Category 15 (Financed Emissions) attribution methodology and Data Quality scores.

What is illustrative in this demonstration

The bank, its branding, branches, customer borrower names where outside the public reference dataset, the loan book, the application queue and the screening recommendations are all synthesized for demonstration. The platform is capable of replacing every illustrative element with real bank source-of-record data once a production integration has been scoped and agreed.

Confidentiality

This manual and the demonstration product it accompanies are shared under the demonstration agreement between First Bank of Nepal and the Jana Earth Data team. Please do not redistribute outside the evaluation working group without written permission. Thank you.

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- 1.1 Audience and intended use
- 1.2 Dashboard layout
- 1.3 Authentication and access
- 1.4 Conventions used in this manual

Part 2 Loan Book tab

- 2.1 Purpose
- 2.2 Filters and search
- 2.3 Sorting and pagination
- 2.4 Borrower detail drilldown

Part 3 ESRM Screening tab

- 3.1 Purpose and regulatory context
- 3.2 Application queue
- 3.3 Screening workbench
- 3.4 ESDD checklist drawer
- 3.5 Sanity check drawer

Part 4 Green Finance Taxonomy tab

- 4.1 Purpose and regulatory context
- 4.2 Taxonomy color codes
- 4.3 Distribution by count and value
- 4.4 Sector view

Part 5 NSRS Disclosure tab

- 5.1 Purpose and regulatory context
- 5.2 Financed emissions headline
- 5.3 Portfolio funnel
- 5.4 Data quality distribution
- 5.5 Multi-year trend
- 5.6 Top contributors

Part 6 Guided tour mode

Part 7 Glossary

Part 8 Data sources and provenance

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The Financed Emissions Dashboard is intended for use by First Bank of Nepal staff who are responsible for environmental and social risk management, green finance reporting, and forward sustainability disclosure under the Nepal Sustainability Reporting Standards. Specifically, the dashboard is designed around four operating roles:

- **Risk officers** reviewing high-emitting loan applications against the NRB ESRM Guidelines (2018) before approval.
- **Green finance and sustainability team** classifying the existing portfolio against the NRB Green Finance Taxonomy (October 2024).
- **Compliance and disclosure team** preparing the bank's first NSRS-aligned financed emissions disclosure for the 2026 to 2027 reporting cycle.
- **Branch managers and relationship officers** seeking visibility into the emissions profile of customers they manage.

Not a regulatory filing tool

Throughout this manual, where the dashboard supports a regulated process such as NRB ESRM filing or NSRS disclosure, the dashboard supports the preparation of those filings rather than acting as the official submission system. Final filings continue to be produced through the bank's existing regulatory reporting workflow.

1.2 Dashboard layout

The dashboard opens to the Loan Book tab. Four tabs are available across the top of the main view, in this order from left to right:

Tabs, left to right	
Loan Book	Browse the bank's loan portfolio with full filtering, sort, and search. Default landing tab.
ESRM Screening	Review and document the ESRM screening process for new and high-risk loan applications, against NRB 2018 ESRM Guidelines.
Green Finance Taxonomy	Inspect the portfolio's classification under the NRB Green Finance Taxonomy (Oct 2024).
NSRS Disclosure	Review financed emissions and PCAF data quality on a disclosure-ready basis, in NPR.

The top-right of the dashboard carries the tour controls (see Part 6) and an at-a-glance indicator of which data sources are currently live versus snapshotted. When connected to the Jana platform, the Climate TRACE 2024 snapshot is served from a stable internal copy rather than re-fetched from the upstream provider for each request.

1.3 Authentication and access

Production deployments of the dashboard are gated by the bank's single sign-on and use role-based access control. In this demonstration the authentication step is shown but not enforced, and any user can navigate all four tabs. In production, the four operating roles listed in section 1.1 each receive a tailored subset of tabs and actions.



Throughout this manual the following conventions are used:

- **(i) icons** in the dashboard show inline tooltips that explain PCAF scores, taxonomy color codes, ESRM risk classifications, and air quality bands. The same explanations are reproduced in Part 7 of this manual.
- **Bold UI labels** in this manual refer to controls, tabs, or buttons that appear in the dashboard itself.
- **NPR** values throughout the NSRS tab follow IFRS S1 presentation-currency guidance. USD appears where it is needed for the PCAF attribution calculation.
- **Demo Purposes Only** appears on every page of this manual and at the bottom of the dashboard application itself.

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The Loan Book tab is the bank's operating view of its full lending portfolio with environmental risk and financed emissions overlays applied. The demonstration portfolio shown contains 80,035 active or under-review loans across retail, SME, commercial, and corporate business units. In a production deployment the same view operates directly against the bank's core banking system of record.

2.2 Filters and search

Filters are displayed across the top of the loan table. They are additive: applying more than one filter narrows the visible set to loans that satisfy all selected filters at once.

Available filters	
Free text	Matches against loan identifier, borrower name, branch name, and sector. Typing in the search field updates the table as you type.
Business unit	Retail, SME, Corporate, or Project Finance.
Status	Active, Disbursed, Under Review, Approved, or Declined. Under Review loans also appear in the ESRM application queue (Part 3).
Taxonomy	Green, Amber, Red, or Unclassified per the NRB Green Finance Taxonomy (October 2024). See Part 4.
Branch	Filter to a specific branch. Useful for branch managers.
Sector	NRB sector classification of the borrower (e.g. Cement and Cement Products, Hydropower, Hotels and Tourism).
Outstanding NPR range	Minimum and maximum outstanding balance in NPR.

2.3 Sorting and pagination

Click any sortable column header to sort the visible result set. By default the loan table is sorted by outstanding NPR balance, largest to smallest. The footer pagination control reports the result set in the form *Showing X to Y of Z loans, page A of B*. Note that Z is the total number of loans matching the current filters, not the total number of loans the bank holds; clearing all filters returns Z to the full portfolio size of 80,035.

2.4 Borrower detail drilldown

Clicking a loan row opens the borrower detail panel. The panel shows the borrower's NRB sector, estimated enterprise value used for PCAF attribution, ownership chain where available, and the list of facilities matched to the borrower. For cement borrowers the facility list is enriched with capacity figures from the Global Cement and Concrete Tracker. The borrower detail panel is the right place to confirm the provenance of any single attribution figure shown elsewhere in the dashboard.

The ESRM tab supports compliance with the NRB Environmental and Social Risk Management Guidelines (2018), specifically the steps an originating branch or risk officer is required to complete before a new loan is approved or a renewal is recommended. Under the NRB ESRM Guidelines, the bank is required to perform and document an Environmental and Social Due Diligence (ESDD) on applicable applications. The ESRM tab structures that work into a consistent screening record. The NRB does not receive every individual ESDD record; rather, the bank reports aggregate ESRM metrics through the regular NRB reporting channels on a periodic basis.

3.2 Application queue

The left-hand application queue lists loans whose status is *under-review*, sorted by attributed CO₂-equivalent emissions in descending order. Selecting a row populates the screening workbench to the right with that borrower's details. The queue is designed to surface the highest emissions exposure first so that risk officers can prioritise the most material reviews.

3.3 Screening workbench

The screening workbench is the core decision-support surface for the ESRM tab. For the selected borrower it presents:

- **Borrower identification:** legal name, NRB sector classification, ownership chain, and headquarters location.
- **Risk classification:** low, medium, high, or extreme, with a color-coded badge and a written rationale.
- **Recommendation:** approve, approve with conditions, or decline, with the reasoning the dashboard surfaces. The final credit decision is always made by the bank's authorised credit committee, not by the dashboard.
- **National emissions share:** the percentage of Nepal's estimated national CO₂ footprint represented by the borrower's matched facilities.
- **Sector benchmark:** the borrower's intensity compared to the EDGAR-derived sector average.
- **Air quality nearby:** the nearest OpenAQ monitoring station and a banded PM_{2.5} reading, where available. The OpenAQ network in Nepal is sparse, so for many facilities no nearby reading is available; the dashboard explicitly says so rather than fabricating a value.

Risk classification colors used throughout this tab are:

	Low	Standard underwriting. No additional environmental conditions.
	Medium	Approve with documented environmental conditions; quarterly monitoring review.
	High	Approve only with strong mitigation commitments and a board-level credit decision.
	Extreme	Recommend decline. Where the bank elects to proceed, escalate to the ESG committee for formal exception.

3.4 ESDD checklist drawer

The **ESDD Checklist** button opens a slide-over drawer that walks the screening officer through the 10 categories of NRB ESRM Annex 5. For each category the drawer captures a status (*complete*, *pending*, *not applicable*), a free-text note, and the supporting evidence reference. The drawer is designed to be completed during the credit committee preparation rather than retrospectively. The 10 ESDD categories follow the NRB Annex 5 ordering and cover: environmental impact, social impact, labour and working conditions, land acquisition, community health and safety, indigenous people, cultural heritage, biodiversity, pollution prevention, and grievance mechanisms.

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The **Sanity Check** button opens a slide-over drawer that documents how the borrower's emissions figure was derived and compares it against an independent reference calculation. For cement borrowers, for example, the drawer shows the Climate TRACE 2024 facility figure alongside an independent calculation from production capacity multiplied by an industry-standard tCO₂ per tonne of cement emission factor. Where the two figures agree to within an order of magnitude, the dashboard treats the Climate TRACE figure as the operating number. Where they diverge, the dashboard flags the divergence and asks the officer to confirm before proceeding.

The Green Finance Taxonomy tab classifies the bank's portfolio against the NRB Green Finance Taxonomy (October 2024). The classification supports the bank's reporting on green lending share, its green bond eligibility analysis, and the alignment view that NSRS disclosure requires.

4.2 Taxonomy color codes

Every loan in the portfolio is assigned one of four taxonomy colors. The (i) tooltip on each color in the dashboard explains the classification rule for that color in the context of the borrower's sector and the use of proceeds. The summary is:

 Green	Use of proceeds is explicitly listed in the NRB green taxonomy. Examples include hydropower under 10 MW, solar, and energy efficiency retrofits in commercial buildings.
 Amber	Transitional activity: the borrower's primary sector is on a credible decarbonisation pathway but is not currently green. Examples include cement plants investing in waste heat recovery or alternative fuels.
 Red	Use of proceeds is explicitly excluded by the NRB taxonomy. Examples include new coal-fired generation, peat extraction, or loans associated with deforestation in protected areas.
 Unclassified	Use of proceeds is unclear or the borrower's documentation does not yet support classification. Unclassified loans should be reviewed during the next renewal.

4.3 Distribution by count and value

The taxonomy summary cards report the portfolio split by both loan count and outstanding NPR balance. The two views can differ materially: a portfolio with many small green retail mortgages may look greener by count than by NPR balance. Both numbers are presented so that disclosure can use whichever view is required.

4.4 Sector view

Below the headline cards the dashboard breaks down each NRB sector by taxonomy color. This view is the operating tool for the green finance team when they are deciding which sectors to prioritise for taxonomy-aligned origination. The (i) tooltips on individual cells explain why a given loan landed in the color it did, so the green finance team can address misclassifications directly with the originating branch.

The NSRS tab is structured for the bank's first NSRS-aligned financed emissions disclosure under the Nepal Sustainability Reporting Standards, which apply to commercial banks for the 2026 to 2027 reporting cycle. The tab follows the PCAF Global GHG Accounting and Reporting Standard, Category 15 (Financed Emissions). All figures are presented in NPR, consistent with the IFRS S1 presentation-currency requirement for sustainability disclosures. Where USD appears it is for the underlying PCAF attribution calculation, which is conventionally performed in USD because borrower enterprise values from public sources are denominated that way.

5.2 Financed emissions headline

The headline figure reports the total attributed CO₂e for the current reporting period and the year-on-year change relative to the prior period. The change indicator explicitly labels the years compared (e.g. 2023 to 2024) to avoid ambiguity. The headline is a sum across in-scope loans only; out-of-scope loans are listed separately so the disclosure remains transparent.

5.3 Portfolio funnel

The portfolio funnel chart walks the reader from total loans, to in-scope loans, to facility-matched loans, to facility-matched unique borrowers. The funnel is the bank's transparency tool: it makes visible how much of the portfolio is supported by the highest-quality PCAF data tier and how much sits on sector benchmarks or revenue-based estimates.

5.4 Data quality distribution

PCAF data quality is reported on a 1 to 5 scale, where 1 is the best (verified emissions reported by the borrower) and 5 is the weakest (revenue-based estimate using sector averages). The distribution chart on the NSRS tab reports the share of the portfolio attributed at each PCAF score, weighted by outstanding NPR balance.

	Score 1	Verified emissions, reported directly by the borrower.
	Score 2	Facility-attributed: Climate TRACE facility data combined with public-filing enterprise value.
	Score 3	Satellite or remote-sensed emissions combined with estimated enterprise value.
	Score 4	Sector benchmark: borrower assigned the EDGAR sector intensity average for Nepal.
	Score 5	Revenue-based estimate using national sector averages.

5.5 Multi-year trend

The multi-year trend chart shows the bank's total attributed CO₂e by reporting year, with a stacked breakdown by taxonomy color so the reader can see whether emissions reductions are coming from genuine green growth or from declining red exposure. The most recent year in the chart is marked as partial when the underlying upstream data is not yet complete for that year.

5.6 Top contributors

The Top Contributors panel lists the loans that account for the largest share of the bank's attributed CO₂e in the current reporting period, sorted in descending order. Each row reports the borrower, the outstanding NPR balance, the attribution factor as a percentage, the attributed CO₂e in tonnes, and the PCAF data quality score.

Clicking a row opens the borrower detail panel described in Part 2.4. The Top Contributors panel is the natural anchor for the engagement priorities section of the bank's TPRS narrative.

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The dashboard ships with an optional guided tour mode that walks a new user through the four tabs with synchronised audio narration. The tour is intended for first-time orientation and for internal training sessions; it does not perform any actions on the underlying data.

6.1 Starting the tour

The Play control is in the top-right of the dashboard. Pressing Play begins the tour at the Loan Book tab and advances through the four tabs in order. Pause, resume, previous step, and next step controls are available throughout. The tour automatically selects a representative borrower on the ESRM tab so the narration and the visible figures stay in sync.

6.2 Audio narration

Audio narration is rendered by OpenAI's text-to-speech engine. The narration text is fixed; the user cannot edit it in the UI. Audio playback is local to the browser session and is not recorded or transmitted beyond the rendering call.

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Definitions

Attribution factor	The fraction of a borrower's total emissions allocated to a specific loan, calculated as outstanding loan balance divided by borrower enterprise value (PCAF convention). Capped at 100% in this demonstration.
Climate TRACE	Global facility-level emissions inventory used as the facility-tier data source. The 2024 v5.6 release is the snapshot referenced throughout the demonstration.
EDGAR	Emissions Database for Global Atmospheric Research, produced by the European Commission Joint Research Centre. The v8.1 gridded release, polygon-clipped to Nepal, supports the national denominator and sector benchmarks.
ESDD	Environmental and Social Due Diligence, the structured screening process the NRB ESRM Guidelines (2018) require for applicable loan applications.
ESRM	Environmental and Social Risk Management. In Nepal the term specifically refers to the NRB ESRM Guidelines (2018).
GCCT	Global Cement and Concrete Tracker, published by Global Energy Monitor. The July 2025 release supplies cement plant capacity and ownership metadata.
GEM	Global Energy Monitor, the publisher of the cement, coal, oil, gas, and wind trackers used to enrich borrower data.
NRB	Nepal Rastra Bank, the central bank of Nepal and the prudential regulator for commercial banks.
NSRS	Nepal Sustainability Reporting Standards, applicable to commercial banks for the 2026 to 2027 reporting cycle, aligned with IFRS S1 and S2.
OpenAQ	Open air quality data network. Nepal coverage is sparse and irregular; the dashboard surfaces a reading only where a station is in operating range of the facility.
PCAF	Partnership for Carbon Accounting Financials, publisher of the Global GHG Accounting and Reporting Standard. Category 15 Financed Emissions is the methodology used throughout.
PCAF Data Quality Score	A 1 to 5 score where 1 represents verified borrower-reported emissions and 5 represents a revenue-based estimate using sector averages.
Taxonomy color	Green, amber, red, or unclassified categorisation of a loan under the NRB Green Finance Taxonomy (October 2024).



The reference data referenced throughout this manual is ingested into the Jana platform and served from the platform's own stabilised store rather than re-fetched from upstream providers at request time. The current snapshot versions are:

Reference data inventory

Climate TRACE	Release v5.6 (2024 data). 213 Nepal assets. Total approximately 28.04 Mt CO ₂ e across the country in 2024.
EDGAR	Release v8.1 gridded global CO ₂ . Polygon-clipped to Nepal yielding 1,343 cells. Estimated 18.81 Mt national CO ₂ for the latest available year.
Global Cement and Concrete Tracker	July 2025 release. 61 operating cement plants in Nepal with capacity, ownership, and geographic detail.
Global Energy Monitor entity registry	Used to resolve ownership chains for cement, hydropower, and heavy industrial borrowers.
NRB ESRM Guidelines	2018 release. Annex 5 ESDD categories drive the ESRM tab checklist.
NRB Green Finance Taxonomy	October 2024 release. Drives the Green Finance Taxonomy tab.
PCAF Global GHG Accounting and Reporting Standard	Latest published edition. Category 15 financed emissions methodology and Data Quality scoring.
OpenAQ	Continuous open data feed. Coverage in Nepal is sparse; the dashboard does not interpolate.

Replacing demonstration data with bank source-of-record data

When this dashboard moves from demonstration to production for First Bank of Nepal, the synthesized loan book and borrower catalog are replaced by a direct integration with the bank's core banking system. The public reference data layer (Climate TRACE, EDGAR, GCCT, NRB documents) remains the same. The integration approach is described in detail in the companion document *Jana Integration with Bank Systems*, included in the same documentation set as this manual.

Thank you

Thank you for taking the time to review this demonstration. Questions and feedback during the evaluation period are welcome and should be directed to the Jana Earth Data team.